

PROGRAM-7

Deploy a web application to kubernetes using Minikube

Project Structure

Program-7

— nginx-deployment.yaml

Step 1: Start the minikube

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ nano nginx-deployment.yaml
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 22.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...

🚧 Docker is nearly out of disk space, which may cause deployments to fail! (92% of capacity). You can pass '--force' to skip this check.
Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
   Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
🔗 Related issue: https://github.com/kubernetes/minikube/issues/9024

🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: metrics-server, storage-provisioner, default-storageclass

! /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
  ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏁 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Check the status of the minikube

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create a nginx-deployment.yaml

```
GNU nano 6.2 nginx-deployment.y
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 4: Apply nginx-deployment.yaml file

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

Step 5: Check the status

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    1/2     2             1           10s
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP    14h
```

Step 6: Expose the deployment as service

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
```

Step 7: Check the service

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ kubectl get service
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP    14h
nginx-deployment    NodePort    10.96.130.15 <none>        80:30372/TCP 17m
```

Step 8: Access the web application

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30372

 Opening service default/nginx-deployment in default browser...

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-7$ Opening in existing browser session.
```